



NLP Text Classification on AG News:

TF-IDF, DistilBERT, Data Efficiency, and LoRA

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Abstract

We study 4-class news topic classification on AG News. We implement a TF-IDF + Logistic Regression baseline, full DistilBERT fine-tuning, a data-efficiency study at 5/10/25/50/100% of training data, and LoRA parameter-efficient fine-tuning. All results are computed from executed experiments; no numbers are hand-entered. We also report calibration (ECE + temperature scaling) and robustness to character-noise shift.

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1 Dataset

Statistics computed from the loaded data:

- Total samples: 127,600
- Train / Test: 120,000 / 7,600
- Classes: 4 (World, Sports, Business, Sci/Tech)
- Train class counts: 'World': 30000, 'Sports': 30000, 'Business': 30000, 'Sci/Tech': 30000
- Missing values (train/test): 0 / 0
- Duplicate texts (train): 0
- Train/test exact-text overlap: 0
- Train length (words): min=8, mean=37.85, median=37.0, max=177

Validation is a stratified 10% split of the training data only; the official test set is pre-served.

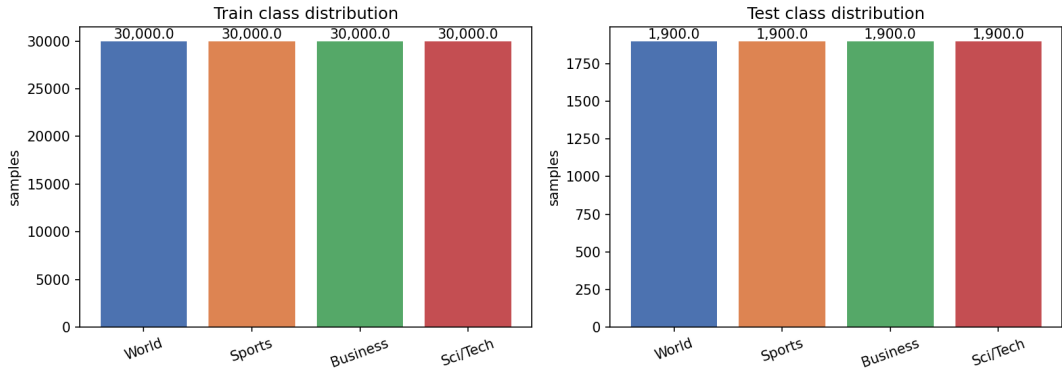


Figure 1: Class distribution.

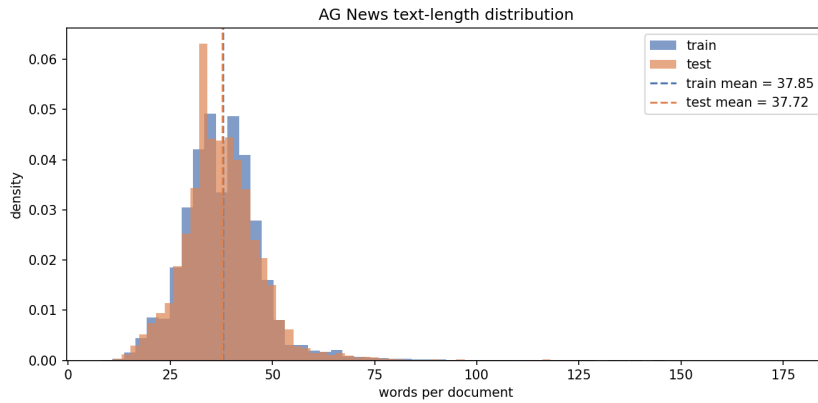


Figure 2: Text-length distribution.

2 Methods

2.1 TF-IDF + Logistic Regression

Unigram+bigram TF-IDF (max features 50,000, min-df 2, sublinear TF) with multinomial Logistic Regression. The vectorizer is fit on the training pool only (leakage-free).

2.2 DistilBERT Full Fine-Tuning

distilbert-base-uncased, max length 128, batch 32, lr 2e-05, 3 epochs, weight decay 0.01, warmup ratio 0.1, fp16, best model by validation macro-F1. Trainable parameters: 66,956,548.

2.3 LoRA

Low-rank adapters on DistilBERT attention projections (q_{lin} , v_{lin}), $r=16$, $\alpha=32$, dropout 0.1. Trainable parameters: 888,580 (1.3097% of the model).

3 Results

3.1 Main comparison

Model	Accuracy	Macro F1	Weighted F1
TF-IDF + LogReg	0.9188	0.9186	0.9186
DistilBERT Full FT	0.9430	0.9430	0.9430
DistilBERT + LoRA	0.9075	0.9073	0.9073

Table 1: Test-set comparison (actual computed values).

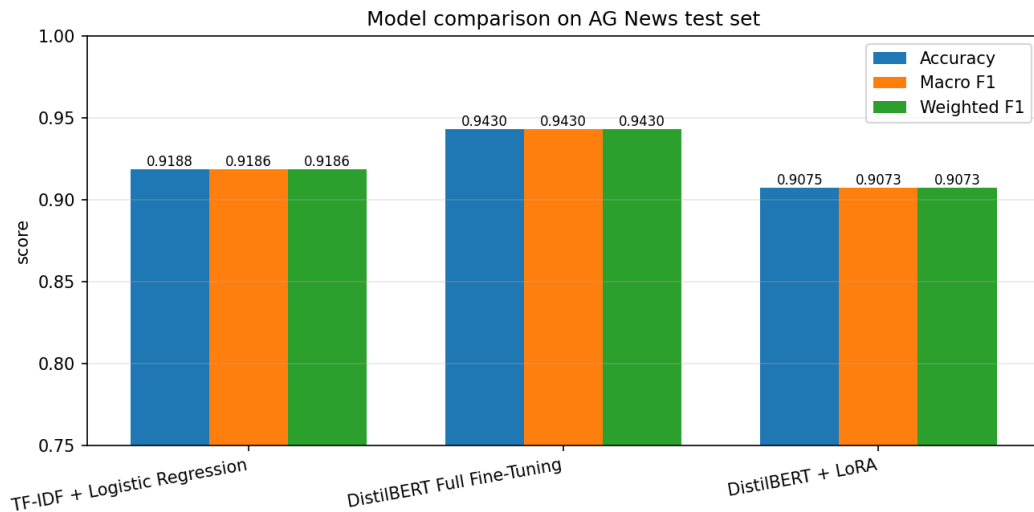


Figure 3: Model comparison.

3.2 Per-class metrics (full fine-tuning)

Class	Precision	Recall	F1	Support
World	0.9571	0.9505	0.9538	1900
Sports	0.9879	0.9884	0.9882	1900
Business	0.9175	0.9074	0.9124	1900
Sci/Tech	0.9100	0.9258	0.9178	1900

Table 2: Per-class metrics, DistilBERT full fine-tuning.

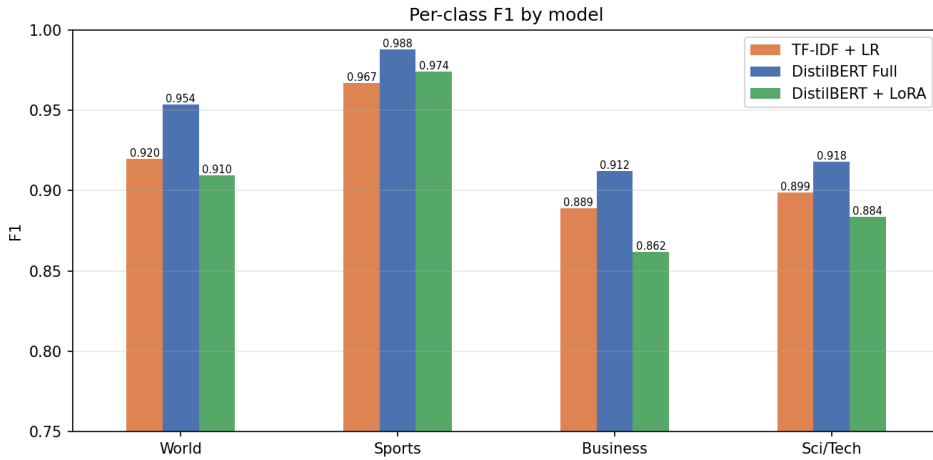


Figure 4: Per-class F1 across models.

3.3 Data efficiency

Fraction	n train	Accuracy	Macro F1	Weighted F1	Time (min)
0.05	5400	0.9008	0.9008	0.9008	2.1
0.10	10800	0.9092	0.9090	0.9090	3.2
0.25	27000	0.9239	0.9239	0.9239	6.5
0.50	54000	0.9337	0.9337	0.9337	12.1
1.00	108000	0.9413	0.9413	0.9413	23.1

Table 3: Data-efficiency results (actual).

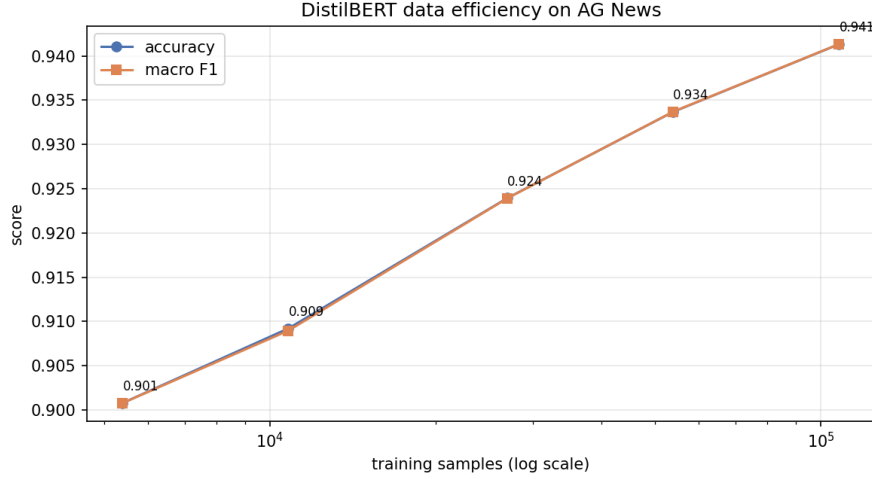


Figure 5: Data-efficiency curve.

3.4 Full fine-tuning vs LoRA

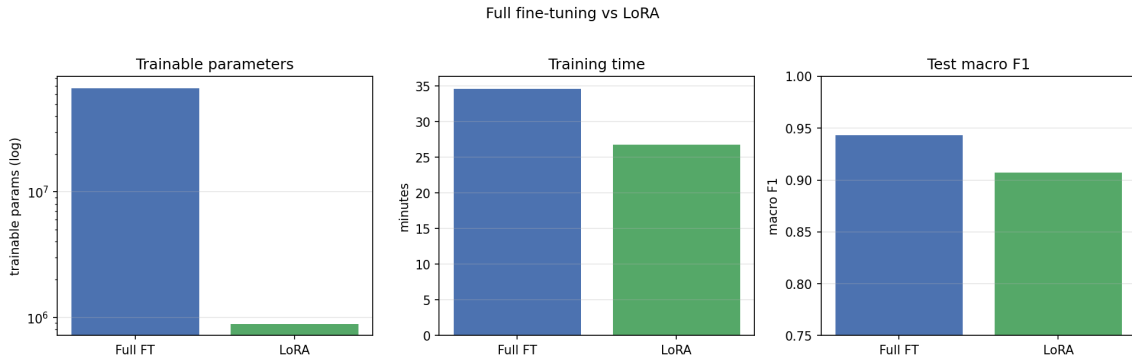


Figure 6: LoRA vs full fine-tuning: parameters, time, performance.

LoRA trains 888,580 parameters vs. 66,956,548 for full fine-tuning (1.33%), at macro-F1 0.9073 vs. 0.9430.

4 Error Analysis

Pair	Count	Row rate
Business → Sci/Tech	131	0.0689
Sci/Tech → Business	102	0.0537
World → Business	46	0.0242
World → Sci/Tech	39	0.0205
Business → World	39	0.0205
Sci/Tech → World	31	0.0163

Table 4: Top confused class pairs.

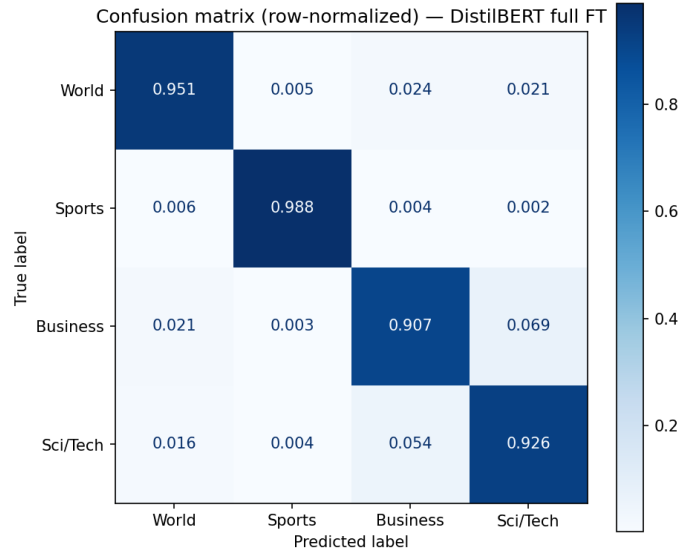


Figure 7: Confusion matrix (row-normalized).

Overall error rate: 0.0570.

Representative misclassifications (highest-confidence errors)

TRUE: Sci/Tech PRED: World conf=0.999
 "Bush Scraps Most U.S. Sanctions on Libya (Reuters) Reuters - President Bush on Monday formally\ended the U.S. trade embargo on Libya to ..."

TRUE: Sci/Tech PRED: Sports conf=0.999
 "Bryant Makes First Appearance at Trial (AP) AP - NBA star Kobe Bryant arrived at his sexual assault trial Monday as attorneys ..."

TRUE: Business PRED: World conf=0.999
 "World briefs LONDON - A man wielding a machete and a knife attacked two security guards at the building housing the headquarters ..."

TRUE: Business PRED: Sports conf=0.999
 "Special to ESPN.com It #39;s the age old question: quot;What do you give to the man who #39;s been everything? quot;. Only ..."

TRUE: Business PRED: World conf=0.999
 "Indonesian diplomats asked to help improve RI #39;s bad image JAKARTA (Antara): President Susilo Yudhoyono asked Indonesian diplomats on Monday to help ..."

TRUE: World PRED: Sports conf=0.999
 "Ljubicic Downs Hanescu at Open De Moselle Top-seeded Ivan Ljubicic of Croatia beat Victor Hanescu of Romania 6-4, 6-4 Tuesday in the ..."

Figure 8: Representative misclassifications.

5 Calibration

Temperature (fit on validation): 1.2085. ECE before / after: 0.0229 / 0.0103. Mean confidence correct vs. incorrect: 0.976 / 0.7989.

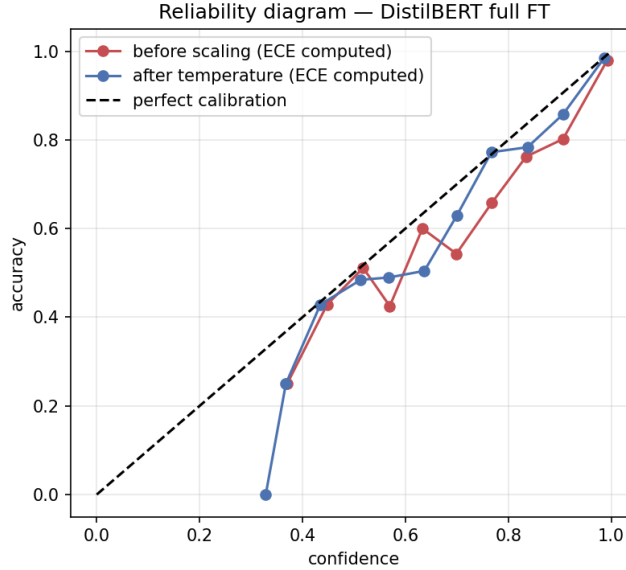


Figure 9: Reliability diagram.

6 Distribution Shift

Character noise $p=0.05$ applied to the test set. Original accuracy $0.9430 \rightarrow 0.9178$; macro-F1 $0.9430 \rightarrow 0.9181$.

7 Convolution vs. Self-Attention

CNNs detect local, position-invariant k -gram patterns via weight sharing and a hard local receptive field; long-range structure needs depth. Self-attention computes content-dependent pairwise weights over all positions in one layer, capturing long-range dependency and context-dependent interactions. See the notebook for the full worked example and hybrid CNN–Transformer designs.

8 Reproducibility and Leakage

All experiments use seed 42. TF–IDF fit on train only; validation from train only; test set touched once per model after decisions; efficiency subsets drawn from the training pool only. Leakage audit passed.

9 Limitations

Results reflect the recorded environment (see notebook). The hybrid CNN–Transformer is discussed analytically, not implemented, to preserve the mandatory experiments’ compute budget.